Quantitative Political Analysis I

Statistical Foundations for Political Inquiry

New College of Florida • Fall 2026

Instructor
Young Joon Oh

Email
yoh@ncf.edu

Office
Announced on Canvas

Office Hours
Tuesday and Thursday, 10:00–11:00 a.m.;
Tuesday, 2:00–3:00 p.m.; and by appointment

Meeting Time
Monday and Wednesday, 11:00 a.m.–12:20 p.m.

Location
See the official course schedule and Canvas

Canvas
Course announcements, readings, data, and guided R files will be posted on the course site

Course Overview

Political arguments are full of numbers. Polls report margins of error. News stories compare outcomes across groups. Researchers ask whether economic conditions, campaign activity, or political institutions are associated with political behavior. Evaluating such claims requires more than recognizing a formula. It requires knowing what was measured, how a statistic was calculated, how much uncertainty surrounds it, and what the evidence does and does not justify.

This course provides a first foundation in statistical reasoning for political inquiry. Students will organize and describe data, reason with probability, understand sampling and polling, construct confidence intervals, test hypotheses, compare groups, analyze contingency tables, and interpret correlation and simple linear regression. The course prepares students for Quantitative Political Analysis II, which develops multivariate regression and applied analysis in R.

No previous statistics or programming is assumed. Students should be comfortable with basic college algebra, including percentages, negative numbers, exponents, square roots, simple equations, tables, and graphs. Most work is calculator-based. During the final two instructional weeks, students will learn to use R for simple regression in regular class meetings; there is no separate lab. R syntax and software operation are not tested on quizzes or examinations. Statistics and R tutoring is available through the Academic Resource Center.

Required Materials

- **Main textbook:** Diez, David M., Mine Çetinkaya-Rundel, and Christopher D. Barr. *OpenIntro Statistics*, 4th ed. Available free from OpenIntro.
- **Supplementary textbook:** Agresti, Alan. *Statistical Methods for the Social Sciences*, 5th ed. The 4th edition is also acceptable. Schedule citations refer to the 5th edition; students using the 4th edition should follow the corresponding chapter topic.

- **Instructor modules, data, and R scripts:** posted on Canvas as assigned.
- **Calculator:** a non-programmable scientific calculator, such as the TI-30XIIS or Casio fx-300ES Plus.
- **R and RStudio:** used for guided simple-regression instruction on November 25 and December 2. Installation instructions and an alternative access option will be provided in advance.

In the schedule, **Main** is required preparation. **Supplement** identifies a selective Agresti reading that adds social-science context or depth; students are not expected to read two complete treatments of the same topic.

Learning Outcomes

By the end of the course, students should be able to:

1. Identify cases, variables, populations, samples, parameters, and statistics, and explain how measurement choices affect a claim.
2. Construct, calculate, and interpret appropriate tables, graphs, and measures of center, spread, and position.
3. Apply basic probability rules and interpret binomial and normal distributions.
4. Explain sampling variability, standard error, sampling distributions, and the Central Limit Theorem.
5. Calculate and interpret confidence intervals and introductory hypothesis tests.
6. Compare means and proportions and analyze relationships in contingency tables.
7. Calculate and interpret correlation and simple linear regression, including guided use of R.
8. Select an appropriate introductory method and evaluate whether the resulting evidence supports a political claim.

Course Format

Class meetings combine explanation, worked calculations, guided practice, and interpretation of statistical claims. On quiz Wednesdays, the quiz occupies 11:00–11:20 a.m.; the remaining 60 minutes are used for a narrower lesson and practice.

Evaluation

Component	Weight
Participation, preparation, and in-class exercises	10%
Quizzes (best 4 of 5)	30%
Midterm examination	25%
Final examination	35%

Participation and In-Class Work

Participation includes attendance, preparation, engagement, and constructive contribution to guided work and discussion. A calculator is required for every quiz and examination. It may be useful during other meetings, but students are not required to bring it unless the instructor announces otherwise.

Because much of the learning occurs through guided practice, regular attendance matters. More than three unexcused absences will substantially reduce the participation grade. Students remain responsible for material and announcements from any missed class.

Quizzes

There will be five 20-minute in-class quizzes; only the best four scores count. Each quiz begins promptly at 11:00 a.m. on Wednesday. A non-programmable scientific calculator is required. Notes, textbooks, and cheat sheets are not permitted. The quizzes are noncumulative: each quiz covers the material completed after the preceding quiz, as shown below.

Quiz	Date	Coverage
Quiz 1	Wednesday, September 2	August 19–31
Quiz 2	Wednesday, September 16	September 2–14
Quiz 3	Wednesday, September 30	September 16–28
Quiz 4	Wednesday, November 18	November 4–16
Quiz 5	Wednesday, December 2	November 18–30

Quiz 5 meets on the last instructional day, one week before the final examination, and doubles as structured review of the regression unit.

A missed quiz ordinarily becomes the dropped quiz. Routine make-up quizzes are not offered.

Midterm and Final Examinations

The midterm examination is scheduled for **Monday, November 2**. The 80-minute examination covers material completed through October 28.

The final examination is cumulative, with greater emphasis on material taught after the midterm. Students must bring **a non-programmable scientific calculator** to both examinations. **One letter-size handwritten cheat sheet** is permitted for each examination. R and R syntax are not tested.

The exact final-examination date, time, and room will be announced on Canvas within the December 9–11 examination period.

New College Course Designations and Narrative Evaluation

New College records a course designation and a narrative evaluation addressing calculation, statistical reasoning, interpretation, method selection, and engagement. The final percentage converts as follows:

Final percentage	NCF designation
90–100%	Strong Satisfactory (Sat+)
80–89.99%	Satisfactory (Sat)
70–79.99%	Marginal Satisfactory (Sat−)
Below 70%	Unsatisfactory (Unsat)

Essential Course Policies

- **Missed examinations:** A missed midterm or final is rescheduled only for a documented illness, family emergency, approved travel, or another circumstance recognized by College policy. Contact the instructor promptly.
- **Academic integrity and AI:** Follow the New College Academic Honor Code. Unless explicitly authorized, AI may not be used for graded work, and submitted calculations, interpretations, and R work must be the student's own.

- **Support and communication:** For accommodations, contact the Accessible Learning Center and the instructor early. Course materials and updates appear on Canvas; check Canvas and New College email regularly.

Course Schedule

Unit 1 — Orientation and the Logic of Political Evidence

Mon, Aug 17

Course orientation

Main: Course syllabus

What QPA I covers • how QPA I leads to QPA II and planned advanced-methods offerings • course materials, assessments, quiz rhythm, calculator rules, and the limited role of R

Wed, Aug 19

Cases, variables, populations, and samples

Main: OpenIntro §§1.2 and 1.3.1
Supplement: Agresti §§1.1–1.2

Cases, observations, variables, and data matrices • categorical and numerical variables • population, sample, parameter, and statistic • description, inference, association, and causal claims

Unit 2 — Measurement and Describing Data

Mon, Aug 24

Concepts, measurement, and coding

Main: Instructor module — From Political Concepts to Variables: Measurement, Coding, Validity, and Reliability
Supplement: Agresti §2.1; previously assigned OpenIntro §1.2 for basic variable types

Concept, variable, value, and operational definition • categorical versus quantitative measurement • nominal, ordinal, interval, and ratio scales • validity, reliability, and measurement error • why assigning numbers to categories does not make every calculation meaningful

Wed, Aug 26

Tables, graphs, and measures of center

Main: OpenIntro §§2.1.2–2.1.3, 2.1.5, and 2.2.1
Supplement: Agresti §§3.1–3.2

Frequency and relative-frequency tables • bar charts and histograms • mean, median, and mode • choosing a display and measure of center from the variable type and distribution

Unit 3 — Spread, Position, and Comparing Distributions

Mon, Aug 31

Variation, position, shape, and outliers

Main: OpenIntro §§2.1.4–2.1.6
Supplement: Agresti §§3.3–3.4

Range, variance, and standard deviation • quartiles, percentiles, and interquartile range • symmetry, skew, outliers, and resistant summaries • choosing summaries that do not conceal the distribution's shape

Wed, Sep 2
Quiz 1
11:00–11:20

Quiz 1 and grouped comparisons

Main: OpenIntro §§2.2.2 and 2.2.6
Supplement: Agresti §3.5

Explanatory and response variables in a grouped description • comparing center, spread, shape, and sample size across groups • conditional percentages for grouped categorical data • why an overall mean can conceal differences between groups

Mon, Sep 7

Labor Day — no class.

Unit 4 — Probability and Conditional Reasoning

Wed, Sep 9

Basic probability

Main: OpenIntro §3.1
Supplement: Agresti §4.1

Sample spaces and events • complements, unions, and intersections • mutually exclusive events • addition rule and the multiplication rule for independent events • moving among counts, proportions, and probabilities

Mon, Sep 14

Conditional probability, independence, and base rates

Main: OpenIntro §3.2

Conditional probability and the importance of the denominator • joint, marginal, and conditional probabilities • general multiplication rule • independent versus dependent events • tree diagrams, reverse-conditional errors, and base rates

Unit 5 — Random Variables and Probability Distributions

Wed, Sep 16
Quiz 2
11:00–11:20

Quiz 2 and random variables

Main: OpenIntro §3.4
Supplement: Agresti §4.2

Discrete and continuous random variables • probability distributions • expected value as a long-run mean • variance and standard deviation of a discrete random variable • checking whether a proposed distribution is valid

Mon, Sep 21

Normal and binomial distributions

Main: OpenIntro §§4.1 and 4.3
Supplement: Agresti §§4.2–4.3

Normal distribution and the 68–95–99.7 rule • z-scores, normal areas, and percentiles • binomial conditions, one exact probability, and an expected count • deciding whether a named distribution is appropriate

Unit 6 — Sampling, Polling, and the Central Limit Theorem

Wed, Sep 23

Sampling and polling

Main: OpenIntro §1.3
Supplement: Agresti §§2.2–2.3

Simple random, stratified, cluster, and convenience sampling • sampling bias, undercoverage, nonresponse, and question wording • random sampling versus random assignment • why a large sample cannot repair biased selection or poor measurement • which population a poll can defensibly describe

Mon, Sep 28

Sampling distributions and the Central Limit Theorem

Main: OpenIntro §5.1; **Instructor module** — Sampling Distributions for Means and Proportions
Supplement: Agresti §§4.4–4.5

A statistic as a random variable across repeated samples • sampling distributions of sample means and sample proportions • standard error and its relationship to sample size • Central Limit Theorem: conditions, intuition, and limits • distinguishing population, sample, and sampling distributions

Unit 7 — Estimation and Confidence Intervals

Wed, Sep 30
Quiz 3
11:00–11:20

Quiz 3 and foundations of estimation

Main: OpenIntro §5.2; previously assigned OpenIntro §5.1
Supplement: Agresti §§5.1–5.2

Estimator, estimate, bias, precision, and standard error • point and interval estimates • critical value and margin of error • confidence interval for one proportion • confidence level, sample size, and interval width

Mon, Oct 5 & Wed, Oct 7

Fall Break — no class.

Mon, Oct 12

Columbus Day/Indigenous People's Day — no class.

Wed, Oct 14

Confidence intervals for proportions and means

Main: OpenIntro §§7.1.1–7.1.4; previously assigned OpenIntro §5.2
Supplement: Agresti §§5.3–5.4

Reactivate estimate $\pm$ critical value $\times$ standard error • conditions for intervals for one proportion and one mean • brief proportion interval and main t-based mean interval • degrees of freedom • correct and incorrect interpretations of 95% confidence

Unit 8 — Hypothesis Testing

Mon, Oct 19

The logic of hypothesis testing

Main: OpenIntro §5.3
Supplement: Agresti §§6.1 and 6.4–6.5

Null and alternative hypotheses • one-sided and two-sided tests • test statistic, p-value, significance level, and decision • Type I and Type II errors • statistical significance, uncertainty, and substantive importance

Wed, Oct 21

One-sample tests for proportions and means

Main: OpenIntro §§6.1.3 and 7.1.5; previously assigned OpenIntro §5.3
Supplement: Agresti §§6.2–6.3

Test for one population proportion • test for one population mean • null-based standard errors • relationship between tests and confidence intervals • collected inference check

Unit 9 — Comparing Two Groups

Mon, Oct 26

Comparing two proportions

Main: OpenIntro §6.2
Supplement: Agresti §§7.1–7.2

Difference in proportions • pooled versus unpooled standard errors • confidence interval and significance test • statistical versus substantive significance • collected method decision

Wed, Oct 28

Comparing two means and recognizing paired data

Main: OpenIntro §§7.2–7.3
Supplement: Agresti §§7.3–7.4

Independent versus paired observations • difference between independent means • paired comparison as one-sample analysis of differences • confidence interval and introductory test • collected design-and-method exit problem

Mon, Nov 2

Midterm examination

The 80-minute midterm covers all material completed through October 28.

Unit 10 — Relationships Between Categorical Variables

Wed, Nov 4

Contingency tables, expected counts, and chi-square

Main: OpenIntro §6.4; previously assigned OpenIntro §2.2
Supplement: Agresti §§8.1–8.2

Joint, marginal, and conditional distributions • statistical independence in a two-way table • expected counts under independence • cell contributions, chi-square statistic, and degrees of freedom • conditions and limitations

Mon, Nov 9

Pattern, strength, and aggregation

Main: OpenIntro §6.4; **Instructor module** — When Aggregates Reverse: Simpson's Paradox in the `UCBAdmissions` Table
Supplement: Agresti §§8.3–8.4

Standardized residuals and which cells drive a chi-square result • evidence of association versus strength • difference in proportions as a strength measure • aggregation, lurking variables, and Simpson's paradox • why significance may coexist with a weak relationship

Wed, Nov 11 *Veterans Day — no class.*

Unit 11 — Method Selection, Scatterplots, and Correlation

Mon, Nov 16

Choosing a method and reading scatterplots

Main: Instructor module — From Research Question to Statistical Method: A Decision Map for One- and Two-Variable Problems; OpenIntro §8.1.1

Identify the observational unit and target parameter • choose among one-sample, two-group, chi-square, correlation, and regression methods • independent versus paired observations • description, association, prediction, and causation • scatterplot direction, form, strength, and unusual observations • R access check

Wed, Nov 18
Quiz 4
11:00–11:20

Quiz 4 and correlation

Main: OpenIntro §8.1.4; **Instructor module** — Same Correlation, Different Evidence: Anscombe's Four Data Sets
Supplement: Agresti §§9.1 and 9.4

Pearson's correlation, introduced conceptually as standardized covariance • direction, strength, units, and the range of r • calculator-based correlation from supplied intermediate sums • outliers, restricted range, nonlinearity, and aggregation • causal caution

Unit 12 — Simple Regression I and Guided R Use

Mon, Nov 23

The least-squares line

Main: OpenIntro §§8.2.1–8.2.7
Supplement: Agresti §§9.2–9.4

Explanatory and outcome variables • least-squares criterion • slope and intercept • fitted values and residuals • calculating a regression line from means, standard deviations, and r • prediction, interpolation, extrapolation, and causal limits

Wed, Nov 25

Simple Regression with R I

Main: Instructor module — Simple Regression with R I: From a CSV to a Scatterplot and Regression

Open a pre-cleaned CSV in RStudio • inspect variable names and units • create a scatterplot with supplied code • use guided templates for `cor()` and `lm()` • connect the coefficients to the line, prediction, and residual concepts learned on Monday

Unit 13 — Simple Regression II, Uncertainty, and Guided R Use

Mon, Nov 30

Regression uncertainty and diagnostics

Main: OpenIntro §§8.2.3, 8.2.7, and 8.3–8.4

Supplement: Agresti §§9.5–9.6

Conditions for simple linear regression • outliers, leverage, influence, and residual patterns • R^2 as proportional reduction in squared error • standard error, confidence interval, and p-value for a slope • prediction versus explanation and association versus causation

Wed, Dec 2

Quiz 5
11:00–11:20

Quiz 5, Simple Regression with R II, and an evidence audit

Main: Instructor module — Simple Regression with R II: Coefficients, Uncertainty, and Residuals

Read coefficient estimates, standard errors, p-values, and R^2 from `lm()` output • produce and interpret a residual plot using supplied code • compare R output with a calculator-based interval • write a complete claim: estimate, uncertainty, substantive meaning, and limitation • identify what additional variables or design would be needed for a causal claim

Final Examination Period

Mon–Tue, Dec 7–8 *Reading/Hurricane Makeup Days — no regular class unless the College announces a required makeup arrangement.*

Wed–Fri, Dec 9–11 *Final examinations. The exact QPA I examination date, time, and room will be announced after the registrar publishes the schedule.*